

Lab 1: Requirements Engineering — Requirements Table

Problem Statement #62 — Food Bank Surplus Redistribution Platform (Sustainability & Green Tech)

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Scope: A web platform where verified Donor Partners (restaurants, grocery stores) list surplus edible food with expiry countdowns, and verified Shelter Coordinators reserve a pickup window and collect the batch using a verification code. A Platform Administrator verifies shelter registrations.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Donor Partner to post a surplus food batch with quantity, a perishable expiry countdown timer and dietary tags (e.g. vegetarian, contains-nuts, halal).	High	Pass: a posted batch appears on the claim board with a live countdown and its dietary tags, and a Shelter Coordinator can reserve it and receive a pickup verification code. Fail: an expired food batch remains active on the claim board.	Core listing functionality; dietary tags and expiry are mandatory for safe redistribution. (Given requirement.)
FR-002	Functional	The system shall enable a Shelter Coordinator to find and filter the claim board based on distance radius, dietary requirements, and the time left until a batch expires.	High	Pass: Applying the filters " within 5 km AND vegetarian " displays only batches that meet both conditions, and the results are shown in under 2 seconds for a board containing 500 active batches . Fail: Any batch that does not match the selected conditions or an expired batch is included in the results.	Shelters have limited time to arrange transportation, so they need to identify suitable food quickly before it expires.
FR-003	Functional	The system shall permit a verified Shelter Coordinator to reserve an available batch for a chosen pickup time before its expiry and generate a unique, single-use pickup verification code for the reservation.	High	Pass: Once confirmed, the batch status changes from AVAILABLE to RESERVED , and a 6-character code is provided to the reserving shelter. If another shelter attempts to reserve the same batch, the request is rejected with an " already claimed " message. Fail: The same batch is successfully reserved by two different shelters.	This prevents multiple shelters from claiming the same physical food batch and provides a verifiable token for the handover.
FR-004	Functional	The system shall allow a Donor Partner to complete a handover by verifying the pickup code supplied by the shelter. After successful verification, the batch status shall become COLLECTED and a timestamp shall be recorded.	Medium	Pass: Providing the correct code changes the status to COLLECTED and records a timestamp in the audit log. An incorrect or previously used code is rejected with an error, and the status remains unchanged. Fail: A batch is marked COLLECTED without a valid verification code.	This confirms that the food was physically handed over to the shelter and supports food-safety traceability and donor reporting.
FR-005	Functional	The system shall automatically remove a batch from the claim board when its expiry countdown reaches zero, prevent any	High	Pass: When a batch reaches T-0 , it is removed from the claim board within 60 seconds . Any attempt to reserve it is rejected, and the donor receives an expiry	Providing food after its expiry can create health risks and legal liability for both the donor and the platform.

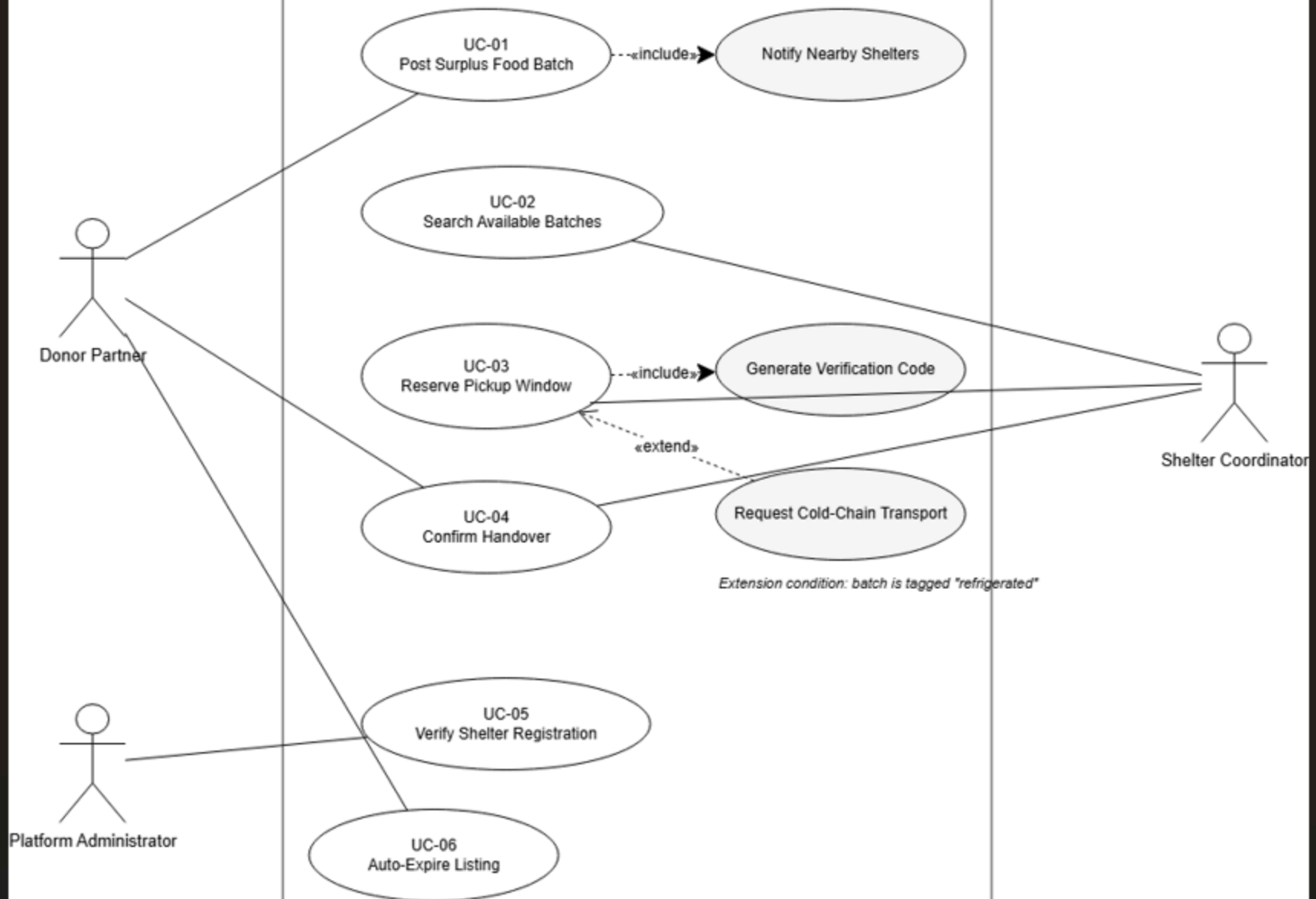
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		further reservations, and inform the donor that the batch has expired.		notification. Fail: An expired batch remains visible on the claim board or can still be reserved..	
NFR-001	Non-functional (Performance & Security)	The platform shall dispatch instant push notifications to Shelter Coordinators within a 5 km radius when a high-quantity perishable food batch is listed, over an authenticated encrypted channel.	High	Pass: benchmarking under simulated peak load (1000 concurrent listings) confirms 95th-percentile notification latency below 10 seconds and that all notification traffic is delivered over TLS 1.2 or higher. Fail: latency or security standards are not met under load.	Perishable food has a very short usable window, so nearby shelters must be alerted immediately. (Given requirement.)
NFR-002	Non-functional (Security & Auditability)	The system shall allow batch reservations only from Shelter accounts that have been approved by a Platform Administrator. It shall also maintain an immutable audit record containing the actor ID, batch ID, action, and timestamp for every listing, reservation, expiry, and collection event.	High	Pass: When an unverified shelter account tries to make a reservation, the system rejects the request with HTTP 403 . In addition, 100% of state transitions during a 50-transaction test run must be recorded in the audit log, with all four fields populated and no record being editable after it is written. Fail: A state change is absent from the audit log, or an unverified account is able to successfully make a reservation.	Food-safety regulations require complete traceability of everyone who handles each batch, while limiting reservations to verified shelters helps prevent donated food from being diverted.

Actors: Donor Partner, Shelter Coordinator, Platform Administrator.

Traceability: FR-001 → UC-01 Post Surplus Food Batch; FR-002 → UC-02 Search Available Batches; FR-003 → UC-03 Reserve Pickup Window; FR-004 → UC-04 Confirm Handover; FR-005 → UC-06 Auto-Expire Listing; NFR-002 → UC-05 Verify Shelter Registration.

UML Use-Case Diagram — Problem Statement #62 *Food Bank Surplus Redistribution Platform*

Food Bank Surplus Redistribution Platform



Lab 1: Use-Case Flow Specification

Use Case UC-03: Reserve Pickup Window

Primary Actor: Shelter Coordinator

Secondary Actor: Donor Partner (receives reservation notification)

Related Requirements: FR-002, FR-003, FR-005, NFR-002

Preconditions

1. The Shelter Coordinator is authenticated and the shelter account has been approved by a Platform Administrator (UC 05).
2. At least one food batch has status AVAILABLE with an expiry countdown greater than zero.
3. The shelter has no unresolved policy block (e.g. repeated no-show on previous pickups).

Postconditions

1. The batch status is RESERVED and is bound to the reserving shelter ID and the chosen pickup window. 2. A unique single-use pickup verification code has been issued to the Shelter Coordinator.
2. The batch no longer appears on the public claim board and cannot be reserved by another shelter. 4. The Donor Partner has been notified, and an immutable audit entry (actor, batch, action, timestamp) has been written.

Main Success Scenario

3. The Shelter Coordinator opens the claim board and applies filters for distance radius, dietary tag and time remaining. 2. The system displays the matching available batches with their live expiry countdowns, quantities and donor locations. 3. The Shelter Coordinator selects a batch and chooses "Reserve".
4. The system re-validates that the batch is still AVAILABLE, unclaimed and has not expired.
5. The system displays the pickup windows that end before the batch expiry deadline.
6. The Shelter Coordinator selects a pickup window and confirms the reservation.
7. The system locks the batch, sets its status to RESERVED against the shelter ID, and generates a unique single-use pickup verification code («include» Generate Verification Code).
8. The system removes the batch from the public claim board and notifies the Donor Partner of the reservation and pickup window.
9. The system displays the verification code and pickup details to the Shelter Coordinator, and writes an audit entry. 10. The use case ends successfully.

Alternate Flow 4a — Batch already reserved by another shelter

- 4a1. The system displays the message "This batch has just been claimed by another shelter."
- 4a2. The system refreshes the claim board and suggests up to three similar batches within the same radius and dietary tag.
- 4a3. The Shelter Coordinator either selects an alternative batch (flow resumes at step 3) or exits. The original batch state is unchanged and no code is issued.

Alternate Flow 6a — Selected pickup window falls after the expiry deadline

6a1. The system rejects the selected window and displays "Pickup must be completed before <expiry timestamp>." 6a2. The system re-displays only the valid windows; the flow resumes at step 6.

Included Use Case: Generate Verification Code — always executed at step 7 («include»).

Extending Use Case: Request Cold-Chain Transport — «extend» at step 6, condition: batch is tagged "refrigerated".